

FYS4480/9480 Sept 10



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$$\overbrace{a_a a_b}^+ = \delta_{ab} \quad a, b > F$$

$$\overbrace{a_p a_q} = \overbrace{a_p^+ a_q^+}$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$a b c d \dots$$

$$F$$

$$|\Phi_0\rangle = |c\rangle$$

$$\vdots$$

$$\vdots$$

$$i j k l \dots$$

$$\vdots$$

$$p q r s \dots$$

$$\sum \langle p | \hat{G}_0 | q \rangle$$

$$p q \times a_p^+ a_q$$

$$a_a^\dagger |c\rangle = a_a^\dagger |\Phi_0\rangle = |\Phi^a\rangle$$

$$\begin{aligned} \overline{\langle c | a_b a_a^\dagger | c \rangle} &= \langle \Phi^b | \Phi^a \rangle \\ &= \delta_{ab} \end{aligned}$$

$$a_j |\Phi_0\rangle = a_j |c\rangle = |\Phi_j\rangle$$

$$j \leq F$$

$$j \in \{\Phi_0\}$$

$|0\rangle$

$$\overline{a_p^+ a_q^-} = 0$$

$$\overline{a_p^+ a_q} = \delta_{pq} n_p$$

$$\overline{a_p a_q^+} = \delta_{pq} (1 - n_p)$$

$$n_p = 1 \quad \text{for } p \leq F$$

$$n_p = 0 \quad \text{for } p > F$$

$$\overline{a_p^+ a_q} \rightarrow \overline{a_a^+ a_b} = 0$$

$$\overline{a_{i'}^+ a_j} = \delta_{i'j} \quad i', j \leq F$$

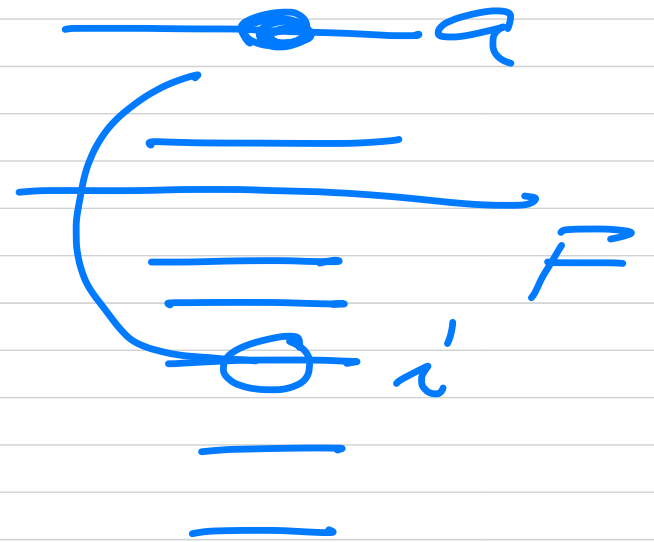
$$\overline{a_a a_b^+} = \delta_{ab} \quad a, b > F$$

$$a_j |\Phi_0\rangle \neq 0$$

$$a_a |\Phi_0\rangle = 0$$

$$a \notin \{\Phi_0\}$$

1p-1h state

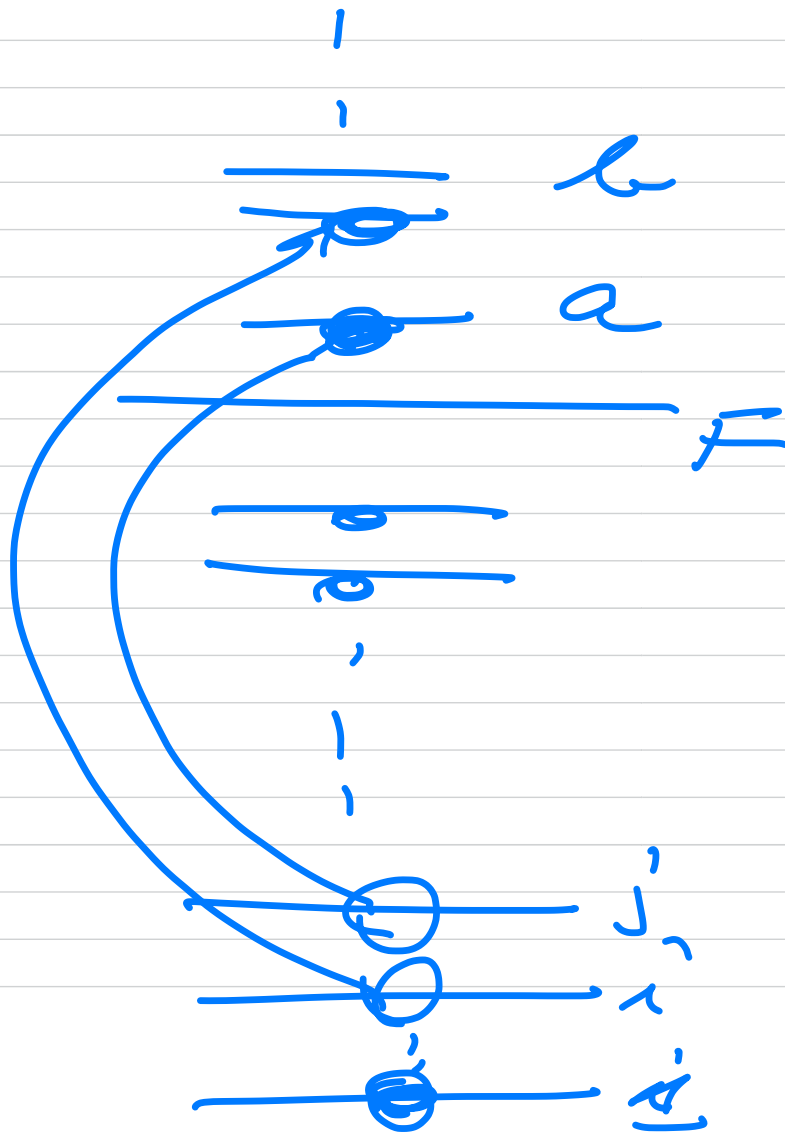


$$a_a^\dagger a_i |\psi\rangle = a_a^\dagger a_i |\Phi_0\rangle$$

$$= |\Phi_i^a\rangle$$

2p2L

$$a_a^+ a_b^+ a_j a_i |\Phi_c\rangle = |\Phi_{ij}^{ab}\rangle$$



$$\overbrace{a_a a_b^\dagger} = \delta_{ab} \quad a, b \in F$$

$$a_{i'}^\dagger a_j = \delta_{i'j} \quad i', j \in F$$

$$\overbrace{a_p a_q} = \overbrace{a_p^\dagger a_q^\dagger} = 0$$

$$\hat{H}_0 = \sum_{pq} \langle p | \hat{H}_0 | q \rangle a_p^\dagger a_q$$

$$= \sum_{i'} \langle i' | \hat{H}_0 | i' \rangle + \sum_{pq} \langle p | \hat{H}_0 | q \rangle \times \{ a_p^\dagger a_q \}$$

$$\hat{H}_I = \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS}$$

$$= \underbrace{\frac{1}{2} \sum_{i'j' \in F} \langle i'j' | v | i'j' \rangle_{AS}}_{\langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle}$$

$$+ \sum_{pq i \in F} \langle pi | v | qi \rangle_{AS} \{ a_p^\dagger a_q \}$$

$$+ \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \{ a_p^\dagger a_q^\dagger a_s a_r \}$$

$$\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$$

$$\begin{aligned} \langle \Phi_0 | \hat{H} | \Phi_0 \rangle &= \langle c | \hat{H} | c \rangle \\ &= \sum_i \underbrace{\langle i | \hat{h}_0 | i \rangle}_{\epsilon_i} + \frac{1}{2} \sum_{i,j} \langle ij | v | ij \rangle_{AS} \end{aligned}$$

$$\hat{F}_N = \sum_{pq} f_{pq} \{ a_p^\dagger a_q \}$$

$$f_{pq} = \langle p | \hat{h}_0 | q \rangle + \sum_i \langle p i | v | q i \rangle$$

$$\hat{V}_N = \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \{ a_p^\dagger a_q^\dagger a_s a_r \}$$

$$\langle \Phi_n^a | \hat{H} | \Phi_0 \rangle = \langle \Phi_n^a | \hat{H} | c \rangle$$

$$\langle \Phi_n^a | E_{\text{ref}} | \Phi_0 \rangle = E_{\text{ref}} \langle \Phi_n^a | \Phi_0 \rangle$$

$$= 0$$

$$\sum_{pq} \langle \Phi_n^a | a_p^\dagger a_q | \Phi_0 \rangle \delta_{pq}$$

$$|\Phi_n^a\rangle = a_q^\dagger a_n | \Phi_0 \rangle$$

$$\langle \Phi_0 | a_n^\dagger \overbrace{a_a a_p^\dagger a_q}^{\delta_{nq} \delta_{ap}} | \Phi_0 \rangle$$

$$= f a_i$$

$$= \langle a | \hat{h}_0 | i \rangle + \sum_j \langle a_j | v | i j \rangle_{A5}$$

$$\langle \Phi_n^q | \hat{V}_N | \Phi_0 \rangle =$$

$$\frac{1}{4} \sum_{p q r s} \langle p q | v | r s \rangle_{A5} = 0$$

$$\times \langle \Phi_0 | \{ a_n^\dagger a_n \} \{ a_p^\dagger a_q^\dagger a_r a_s \} | \Phi_0 \rangle = 0$$

$$\langle \Phi_i^a | \hat{H} | \Phi_0 \rangle$$

$$= f_{ai}$$

$$\langle \Phi_{ij}^{ab} | \hat{H} | \Phi_0 \rangle = ?$$

$$\langle \Phi_i | \hat{H} | \Phi_j \rangle$$

$$|\psi_I\rangle = \sum_j C_{IJ} |\Phi_j\rangle$$